

12 Simulating Randomness

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1 Simulating Randomness and Random Numbers

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1.1 Introduction

This notebook covers the fundamentals of computer simulation, focusing on the essential role of **random number generation**. We'll explore why and how simulations are used, the concept of pseudorandomness, and methods for generating random numbers following various distributions.

A simulation is a computer program that **imitates reality** to study situations and make decisions. We use them when: * Actual experiments are **not feasible** or the system **doesn't exist**. * The **cost** (money, time, danger) of real experiments is **prohibitive**. * We need to **test alternatives** efficiently.

1.2 Element of Chance: Pseudorandom Numbers

At the core of most simulations is **random number generation**. Since a computer follows deterministic instructions, the numbers generated are called **pseudorandom numbers**.

- **Pseudorandom:** Generated by an algorithm (an equation or set of rules), meaning the sequence is entirely predictable if the starting point (the **seed**) is known. They only *appear* to be random.
- **Monte Carlo Simulation:** Any simulation involving a probabilistic model and an element of chance is often referred to as a Monte Carlo simulation.

1.2.1 Multiplicative Linear Congruential Method (MLCM)

The MLCM is a common algorithm for generating a sequence of pseudorandom integers:

$$\text{random}_n = (\text{multiplier} \times \text{random}_{n-1}) \bmod \text{modulus}, \text{ for } n > 0$$

A floating-point number rand in $[0.0, 1.0)$ is then derived by dividing the integer by the modulus:

$$\text{rand} = \frac{\text{random}_n}{\text{modulus}}$$

```
[261]: def mlcm_generator(seed, multiplier, modulus, count):
        """Generates a sequence of pseudorandom integers and floating-point numbers.
        ↪ """
        random_n = seed
        integers = [random_n]

        for _ in range(count - 1):
            random_n = (multiplier * random_n) % modulus
            integers.append(random_n)

        # Convert to floating point numbers in [0.0, 1.0)
        floats = [i / modulus for i in integers]

        return integers, floats

seed = 10
multiplier = 7
modulus = 11
count = 10

integers, floats = mlcm_generator(seed, multiplier, modulus, count)

print(f"Seed: {seed}, Multiplier: {multiplier}, Modulus: {modulus}")
print(f"\nGenerated Integers: {integers}")
print(f"Generated Floats (First 5): {[round(f, 4) for f in floats[:5]]}")
```

Seed: 10, Multiplier: 7, Modulus: 11

Generated Integers: [10, 4, 6, 9, 8, 1, 7, 5, 2, 3]

Generated Floats (First 5): [0.9091, 0.3636, 0.5455, 0.8182, 0.7273]

1.3 Transforming Uniform Random Numbers

The fundamental random number $\text{rand} \in [0.0, 1.0)$ needs to be scaled and shifted to fit the simulation's required range.

1.3.1 1. Random Floating Point Number $\min \leq r < \max$

$$r = (\max - \min) \times \text{rand} + \min$$

1.3.2 2. Random Integer $\min \leq n \leq \max$ (Inclusive)

$$n = \text{int}((\max - \min + 1) \times \text{rand} + \min)$$

```
[262]: import random

# Example 1: Floating Point between 20.0 and 26.3
```

```

min_f = 20.0
max_f = 26.3
rand_f = random.random() # Assumes a new rand in [0.0, 1.0)
r = (max_f - min_f) * rand_f + min_f
print(f"Floating Point ({min_f} to {max_f}): {r:.4f}")

# Example 2: Integer between 20 and 26 (inclusive)
min_i = 20
max_i = 26
rand_i = random.random() # Assumes a new rand in [0.0, 1.0)
n = int((max_i - min_i + 1) * rand_i + min_i)
print(f"Integer ({min_i} to {max_i} inclusive): {n}")

# Python's built-in equivalent
print(f"\nPython's built-in integer method: {random.randint(min_i, max_i)}")

```

Floating Point (20.0 to 26.3): 20.8068

Integer (20 to 26 inclusive): 21

Python's built-in integer method: 24

1.4 Distributions: Discrete vs. Continuous

The **distribution** describes the portion of times each possible outcome occurs.

1.4.1 1. Discrete Distribution (Unequal Probabilities)

To generate random events $e_1, e_2, \dots, e_n$ with specific probabilities $p_1, p_2, \dots, p_n$ (where $\sum p_i = 1$), we use the **cumulative probability** method. We map the uniform $\text{rand} \in [0, 1)$ to a specific event by checking which cumulative range it falls into.

Example: Simulating a coin toss where heads has a 70% chance ($p_H = 0.7$) and tails has a 30% chance ($p_T = 0.3$). * If $\text{rand} < 0.7$ (i.e., p_H), return **Heads**. * Else ($\text{rand} \geq 0.7$), return **Tails**.

```

[263]: def discrete_simulator(events_probs):
        """Generates a random event based on a dictionary of {event: probability}.
        ↪ """
        rand = random.random() # Uniform random number in [0.0, 1.0)
        cumulative_prob = 0.0

        for event, prob in events_probs.items():
            cumulative_prob += prob
            if rand < cumulative_prob:
                return event

        # Fallback (shouldn't happen if probabilities sum to 1)
        return list(events_probs.keys())[-1]

```

```

biased_coin = {"Heads": 0.7, "Tails": 0.3}
results = [discrete_simulator(biased_coin) for _ in range(10000)]

head_count = results.count("Heads")
tail_count = results.count("Tails")

print(f"Simulation of 10,000 biased tosses:")
print(f"Heads Count: {head_count} ({head_count/10000:.2f})")
print(f"Tails Count: {tail_count} ({tail_count/10000:.2f})")

```

Simulation of 10,000 biased tosses:
Heads Count: 6961 (0.70)
Tails Count: 3039 (0.30)

1.5 Continuous Distributions: Advanced Methods

1.5.1 2. Normal (Gaussian) Distribution

This is the famous **Bell Curve**, defined by the mean (μ) and standard deviation (σ). Generating numbers from this distribution often uses the **Box-Muller-Gauss Method**, which converts two uniform random numbers into two independent normally distributed numbers.

We use the `numpy` library in Python for efficiency, which internally uses algorithms like Box-Muller.

```

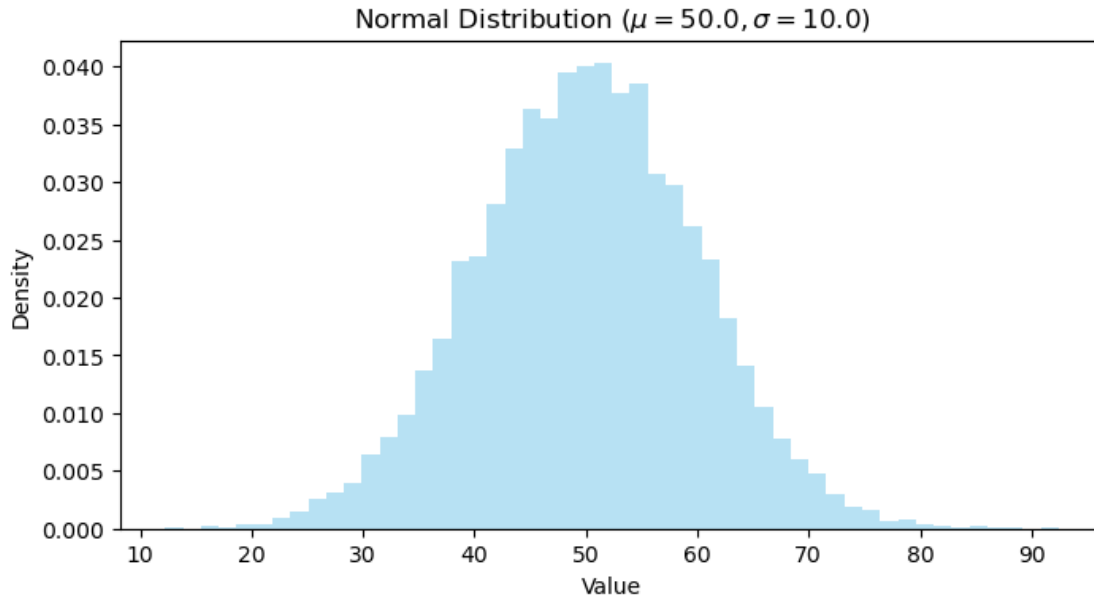
[264]: import numpy as np
import matplotlib.pyplot as plt

# Parameters for the Normal Distribution
mu = 50.0 # Mean (e.g., average test score)
sigma = 10.0 # Standard Deviation (e.g., variability of scores)
num_samples = 10000

# Generate random numbers using a built-in method (based on Box-Muller)
normal_samples = np.random.normal(loc=mu, scale=sigma, size=num_samples)

# Plot the results to visualize the Bell Curve
plt.figure(figsize=(8, 4))
plt.hist(normal_samples, bins=50, density=True, alpha=0.6, color='skyblue')
plt.title(f'Normal Distribution ( $\mu$ ={mu},  $\sigma$ ={sigma})')
plt.xlabel('Value')
plt.ylabel('Density')
plt.show()

```



1.5.2 Manual Implementation: Box-Muller-Gauss Method

While NumPy provides a built-in normal distribution generator, let's manually implement the **Box-Muller algorithm** to understand how it transforms uniform random numbers into normally distributed values.

The Box-Muller Transformation:

$$z_0 = \sqrt{-2 \ln(u_1)} \cos(2\pi u_2)$$

$$z_1 = \sqrt{-2 \ln(u_1)} \sin(2\pi u_2)$$

where u_1, u_2 are independent uniform random numbers in $[0, 1)$.

We compute: - $b = \sqrt{-\ln(\text{rand})}$ - $a = \text{\$ uniform random in } [0, 2\pi)$ - Then: $b \sin(a) + \mu$ and $b \cos(a) + \mu$

To transform to a desired mean μ and standard deviation σ :

$$x = \mu + \sigma \times z$$

```
[265]: def box_muller_normal(mu=0, sigma=1):
        """Generate two independent normal random variables using Box-Muller."""
        # Generate two uniform random numbers
        u1 = random.random()
        u2 = random.random()

        # Avoid log(0)
        while u1 == 0:
```

```

    u1 = random.random()

    # Box-Muller transformation
    b = np.sqrt(-2 * np.log(u1))
    a = 2 * np.pi * u2

    z0 = b * np.cos(a) # First standard normal variable
    z1 = b * np.sin(a) # Second standard normal variable

    # Transform to desired mean and standard deviation
    x0 = mu + sigma * z0
    x1 = mu + sigma * z1

    return x0, x1

# Generate samples using Box-Muller
mu, sigma = 100, 15
samples_bm = []
for _ in range(5000):
    x0, x1 = box_muller_normal(mu, sigma)
    samples_bm.extend([x0, x1])

# Compare with NumPy's built-in
samples_numpy = np.random.normal(mu, sigma, 10000)

# Visualize comparison
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Box-Muller implementation
axes[0].hist(samples_bm, bins=50, density=True, alpha=0.6, color='orange',
             edgecolor='black')
axes[0].set_title(f'Box-Muller Method (={mu}, ={sigma})')
axes[0].set_xlabel('Value')
axes[0].set_ylabel('Density')
axes[0].grid(True, alpha=0.3)

# NumPy implementation
axes[1].hist(samples_numpy, bins=50, density=True, alpha=0.6, color='skyblue',
             edgecolor='black')
axes[1].set_title(f'NumPy Normal (={mu}, ={sigma})')
axes[1].set_xlabel('Value')
axes[1].set_ylabel('Density')
axes[1].grid(True, alpha=0.3)

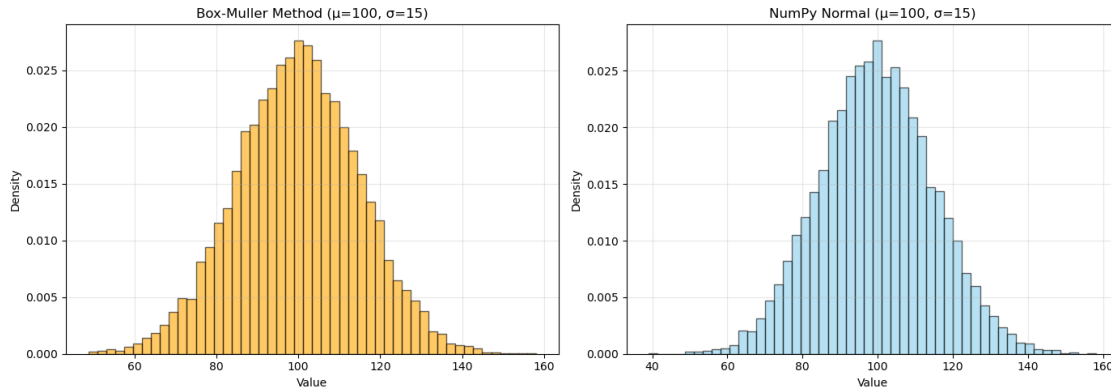
plt.tight_layout()
plt.show()

```

```

print("Box-Muller Implementation:")
print(f" Sample mean: {np.mean(samples_bm):.2f} (Expected: {mu})")
print(f" Sample std: {np.std(samples_bm):.2f} (Expected: {sigma})")
print("\nNumPy Implementation:")
print(f" Sample mean: {np.mean(samples_numpy):.2f} (Expected: {mu})")
print(f" Sample std: {np.std(samples_numpy):.2f} (Expected: {sigma})")

```



Box-Muller Implementation:

Sample mean: 99.86 (Expected: 100)

Sample std: 14.99 (Expected: 15)

NumPy Implementation:

Sample mean: 99.70 (Expected: 100)

Sample std: 15.12 (Expected: 15)

1.5.3 3. Exponential Distribution

Used to model the time between events. It is often generated using the **Inverse Transform Sampling** method, where you solve the inverse of the Cumulative Distribution Function (CDF).

For the Exponential distribution, this results in the formula:

$$t = -\frac{1}{r} \ln(\text{rand})$$

where r is the **rate** parameter (or $1/\lambda$, the mean time between events).

```

[266]: # Example: Simulating the time (t) between calls, where the average time is 5
        ↪ minutes (Rate r = 1/5 = 0.2)
rate_r = 0.2 # The rate parameter
num_samples = 10000

# Implementation of the Inverse Transform Method
rand_samples = np.random.rand(num_samples)

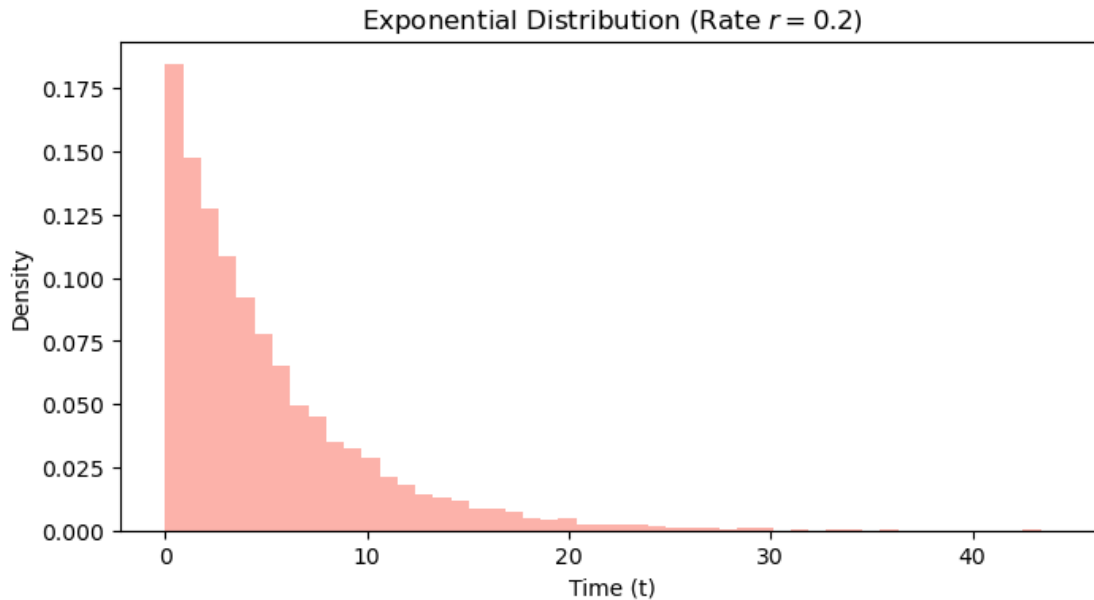
```

```

t_samples = -np.log(rand_samples) / rate_r

plt.figure(figsize=(8, 4))
plt.hist(t_samples, bins=50, density=True, alpha=0.6, color='salmon')
plt.title(f'Exponential Distribution (Rate  $r={rate\_r}$ )')
plt.xlabel('Time (t)')
plt.ylabel('Density')
plt.show()

```



1.6 The Rejection Method

The **Rejection Method** is a general-purpose approach used when direct transformation methods (like the ones above) are not feasible for a complex distribution $f(x)$.

Core Idea: 1. **Select a point** x (from a uniform range $[a, b]$) and a random height y (from $[0, \max(f(x))]$). 2. **Check:** If the height y is below the curve, $y < f(x)$, **accept** x . 3. **Reject:** If $y \geq f(x)$, **reject** x and try again.

This simulates ‘throwing darts’ at the area under the curve and only keeping the darts that hit the target region, thus sampling the desired distribution.

```

[267]: def rejection_method(f, a, b, f_max, max_attempts=10000):
        """
        Generate a random number from distribution  $f(x)$  using the Rejection Method.

        Parameters:

```



```

- f: The probability density function
- a, b: Interval [a, b] for x values
- f_max: Maximum value of f(x) in the interval
- max_attempts: Maximum number of attempts before giving up
"""
for _ in range(max_attempts):
    rand_interval = random.uniform(a, b) # Random x in [a, b]
    rand_upper_bound = random.uniform(0, f_max) # Random y in [0, f_max]

    if rand_upper_bound < f(rand_interval): # Accept if point is under the
↪ curve
        return rand_interval

    raise ValueError("Failed to generate random number within max attempts")

# Example: Custom distribution f(x) = x for x in [0, 2]
def triangular_pdf(x):
    """Triangular distribution: f(x) = x for x in [0, 2]"""
    return x if 0 <= x <= 2 else 0

# Generate samples
samples = [rejection_method(triangular_pdf, 0, 2, 2) for _ in range(5000)]

# Visualize
plt.figure(figsize=(12, 5))

# Plot histogram
plt.subplot(1, 2, 1)
plt.hist(samples, bins=50, density=True, alpha=0.6, color='green',
↪ edgecolor='black')
plt.title('Rejection Method: Triangular Distribution')
plt.xlabel('Value')
plt.ylabel('Density')
plt.grid(True, alpha=0.3)

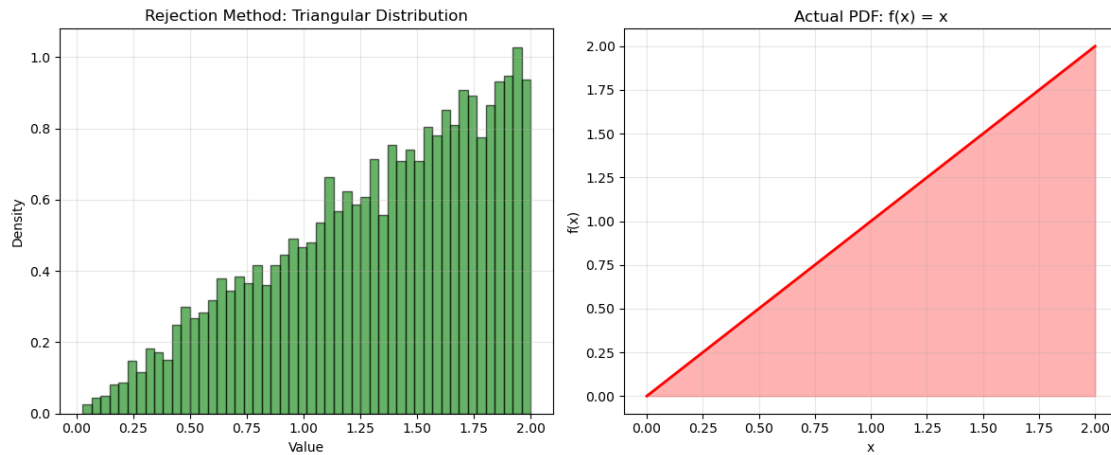
# Plot the actual Probability Density Function (PDF)
x_vals = np.linspace(0, 2, 100)
y_vals = [triangular_pdf(x) for x in x_vals]
plt.subplot(1, 2, 2)
plt.plot(x_vals, y_vals, 'r-', linewidth=2)
plt.fill_between(x_vals, y_vals, alpha=0.3, color='red')
plt.title('Actual PDF: f(x) = x')
plt.xlabel('x')
plt.ylabel('f(x)')
plt.grid(True, alpha=0.3)

plt.tight_layout()

```

```
plt.show()

print(f"Generated {len(samples)} samples using Rejection Method")
print(f"Mean: {np.mean(samples):.4f} (Expected: ~1.33)")
print(f"Std Dev: {np.std(samples):.4f}")
```



Generated 5000 samples using Rejection Method
Mean: 1.3288 (Expected: ~1.33)
Std Dev: 0.4722

1.7 Monte Carlo Simulation: Estimating

One of the most famous **Monte Carlo simulations** is estimating the value of π by “throwing darts” at a square containing an inscribed circle.

1.7.1 The Method

- **Square** has side length $2r$, so area = $(2r)^2 = 4r^2$
- **Circle** has radius r , so area = πr^2
- The ratio of areas: $\frac{\pi r^2}{4r^2} = \frac{\pi}{4}$
- Therefore: $\pi \approx 4 \times \frac{\text{points in circle}}{\text{total points}}$

1.7.2 Algorithm

1. Generate random points (x, y) in the square $[-r, r] \times [-r, r]$
2. Check if each point is inside the circle: $x^2 + y^2 < r^2$
3. Calculate the ratio and multiply by 4 to estimate

```
[268]: def estimate_pi(num_points):
        """Estimate using Monte Carlo method."""
        radius = 1.0
```

```

points_in_circle = 0

for _ in range(num_points):
    # Generate random point in square [-radius, radius] x [-radius, radius]
    x = random.uniform(-radius, radius)
    y = random.uniform(-radius, radius)

    # Check if point is inside circle:  $x^2 + y^2 < r^2$ 
    if x**2 + y**2 < radius**2:
        points_in_circle += 1

pi_estimate = 4 * points_in_circle / num_points
return pi_estimate

# Test with increasing number of points
trials = [100, 1000, 10000, 100000, 1000000]
estimates = []
errors = []

print("Monte Carlo Estimation of :")
print("=" * 55)
for n in trials:
    est = estimate_pi(n)
    estimates.append(est)
    error = abs(est - np.pi)
    errors.append(error)
    print(f"Points: {n:>8,} |      {est:.6f} | Error: {error:.6f}")

print(f"\nActual : {np.pi:.10f}")

```

```

Monte Carlo Estimation of :
=====
Points:      100 |      2.960000 | Error: 0.181593
Points:     1,000 |      3.016000 | Error: 0.125593
Points:    10,000 |      3.148400 | Error: 0.006807
Points:   100,000 |      3.148640 | Error: 0.007047
Points: 1,000,000 |      3.141344 | Error: 0.000249

Actual : 3.1415926536
Points: 1,000,000 |      3.141344 | Error: 0.000249

Actual : 3.1415926536

```

```

[269]: from matplotlib.patches import Circle, Rectangle

# Visualize the convergence and dart-throwing process
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

```

```

# Plot 1: Estimates vs Number of Points
axes[0].plot(trials, estimates, 'bo-', label='Estimates', linewidth=2,
    ↪markersize=8)
axes[0].axhline(y=np.pi, color='r', linestyle='--', linewidth=2, label=f'Actual
    ↪ = {np.pi:.6f}')
axes[0].set_xscale('log')
axes[0].set_xlabel('Number of Points', fontsize=11)
axes[0].set_ylabel('Estimated ', fontsize=11)
axes[0].set_title('Monte Carlo Convergence to ', fontsize=12,
    ↪fontweight='bold')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

# Plot 2: Visual representation of the dart-throwing
num_vis_points = 2000
radius = 1.0
x_points = [random.uniform(-radius, radius) for _ in range(num_vis_points)]
y_points = [random.uniform(-radius, radius) for _ in range(num_vis_points)]

# Classify points
inside = [(x, y) for x, y in zip(x_points, y_points) if x**2 + y**2 < radius**2]
outside = [(x, y) for x, y in zip(x_points, y_points) if x**2 + y**2 >=
    ↪radius**2]

# Plot
if inside:
    axes[1].scatter(*zip(*inside), c='red', s=1, alpha=0.5, label='Inside
    ↪Circle')
if outside:
    axes[1].scatter(*zip(*outside), c='blue', s=1, alpha=0.5, label='Outside
    ↪Circle')

# Draw circle
circle = Circle((0, 0), radius, fill=False, color='black', linewidth=2)
axes[1].add_patch(circle)

# Draw square
square = Rectangle((-radius, -radius), 2*radius, 2*radius,
    fill=False, color='green', linewidth=2)
axes[1].add_patch(square)

axes[1].set_xlim(-1.2, 1.2)
axes[1].set_ylim(-1.2, 1.2)
axes[1].set_aspect('equal')

```

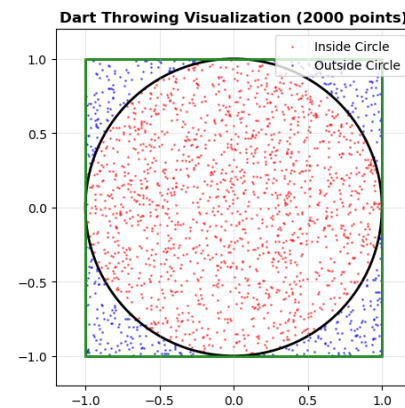
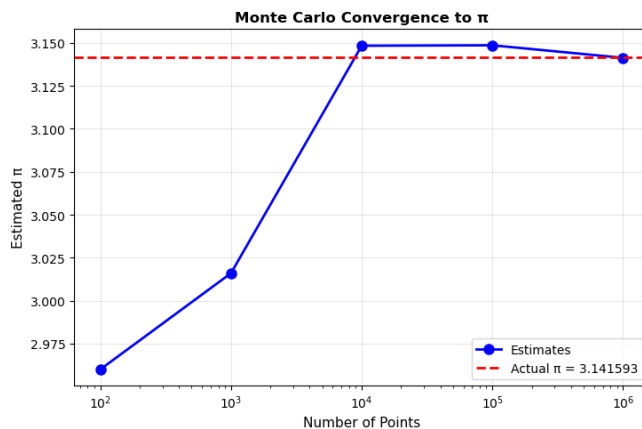
```

axes[1].set_title(f'Dart Throwing Visualization ({num_vis_points} points)',
    ↪fontsize=12, fontweight='bold')
axes[1].legend(loc='upper right')
axes[1].grid(True, alpha=0.3)

plt.tight_layout()
plt.show()

pi_from_vis = 4 * len(inside) / num_vis_points
print(f"\nFrom visualization ({num_vis_points} points):    {pi_from_vis:.6f}")

```



From visualization (2000 points): 3.114000

1.8 Comparing Different Distributions

Let's visualize the four main distribution types side-by-side to understand their characteristics: 1. **Uniform** - Equal probability across the entire range 2. **Normal (Gaussian)** - Bell curve, most common in nature 3. **Exponential** - Models time between events in Poisson processes 4. **Discrete** - Specific outcomes with defined probabilities (e.g., dice roll)

```

[270]: fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# 1. Uniform Distribution
uniform_samples = np.random.uniform(0, 10, 10000)
axes[0, 0].hist(uniform_samples, bins=50, density=True, alpha=0.7,
    ↪color='blue', edgecolor='black')
axes[0, 0].set_title('Uniform Distribution [0, 10]', fontsize=12,
    ↪fontweight='bold')
axes[0, 0].set_xlabel('Value')
axes[0, 0].set_ylabel('Density')

```

```

axes[0, 0].grid(True, alpha=0.3)

# 2. Normal Distribution
normal_samples = np.random.normal(50, 10, 10000)
axes[0, 1].hist(normal_samples, bins=50, density=True, alpha=0.7,
    color='green', edgecolor='black')
axes[0, 1].set_title('Normal Distribution ( =50, =10)', fontsize=12,
    fontweight='bold')
axes[0, 1].set_xlabel('Value')
axes[0, 1].set_ylabel('Density')
axes[0, 1].grid(True, alpha=0.3)

# 3. Exponential Distribution
exponential_samples = np.random.exponential(5, 10000)
axes[1, 0].hist(exponential_samples, bins=50, density=True, alpha=0.7,
    color='red', edgecolor='black')
axes[1, 0].set_title('Exponential Distribution (mean=5)', fontsize=12,
    fontweight='bold')
axes[1, 0].set_xlabel('Value')
axes[1, 0].set_ylabel('Density')
axes[1, 0].grid(True, alpha=0.3)

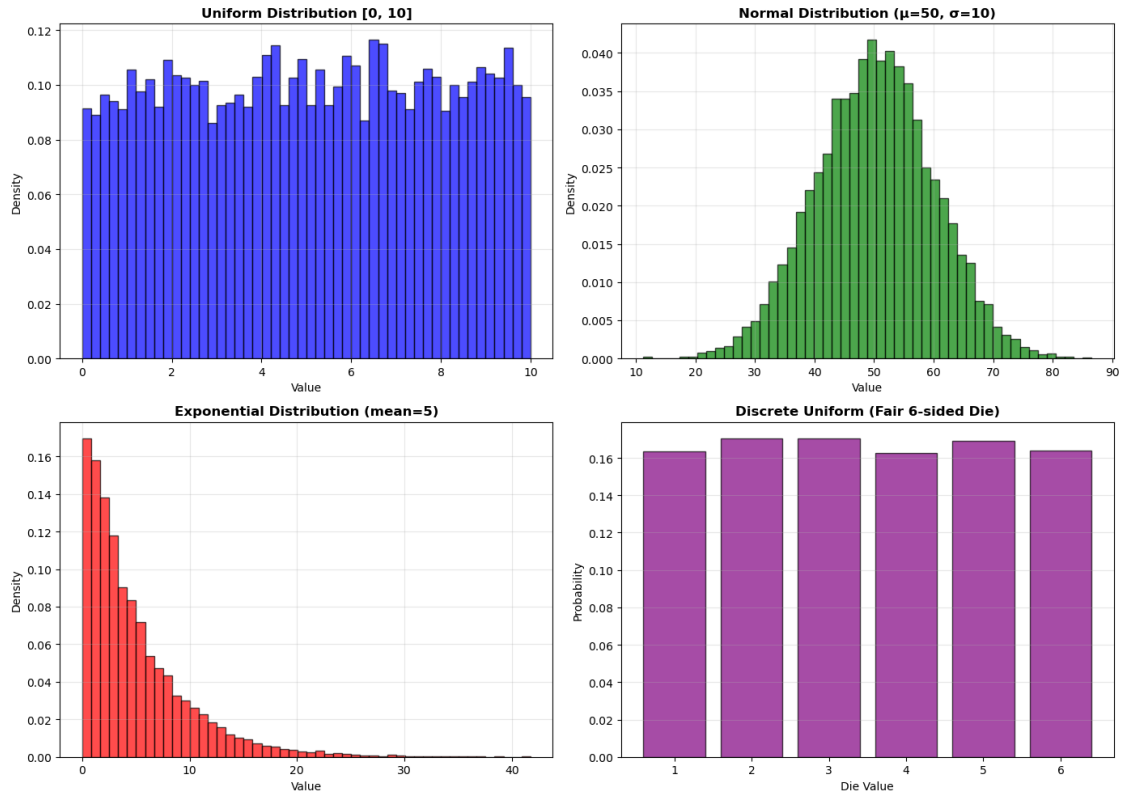
# 4. Discrete Distribution (Fair Dice)
dice_rolls = [random.randint(1, 6) for _ in range(10000)]
axes[1, 1].hist(dice_rolls, bins=np.arange(0.5, 7.5, 1), density=True,
    alpha=0.7, color='purple', edgecolor='black', rwidth=0.8)
axes[1, 1].set_title('Discrete Uniform (Fair 6-sided Die)', fontsize=12,
    fontweight='bold')
axes[1, 1].set_xlabel('Die Value')
axes[1, 1].set_ylabel('Probability')
axes[1, 1].set_xticks(range(1, 7))
axes[1, 1].grid(True, alpha=0.3, axis='y')

plt.tight_layout()
plt.show()

# Print statistics
print("Distribution Statistics (10,000 samples):")
print("=" * 60)
print(f"Uniform:      Mean = {np.mean(uniform_samples):.2f}, Std = {np.
    std(uniform_samples):.2f}")
print(f"Normal:        Mean = {np.mean(normal_samples):.2f}, Std = {np.
    std(normal_samples):.2f}")
print(f"Exponential:   Mean = {np.mean(exponential_samples):.2f}, Std = {np.
    std(exponential_samples):.2f}")

```

```
print(f"Discrete:      Mean = {np.mean(dice_rolls):.2f}, Std = {np.
      ↪std(dice_rolls):.2f}")
```



Distribution Statistics (10,000 samples):

```
=====
Uniform:      Mean = 5.05, Std = 2.87
Normal:       Mean = 50.15, Std = 9.97
Exponential:  Mean = 5.00, Std = 4.92
Discrete:     Mean = 3.50, Std = 1.70
```

1.9 Summary and Key Takeaways

1.9.1 Random Number Generation Methods

Distribution	Generation Method	Formula/Algorithm
Uniform	Direct scaling	$r = (\max - \min) \times \text{rand} + \min$
Discrete	Cumulative probability	Map rand to cumulative probability ranges
Normal	Box-Muller-Gauss	$b \sin(a) + \mu$ where $b = \sqrt{-\ln(\text{rand})}$, $a \in [0, 2\pi)$
Exponential	Inverse Transform	$t = -\frac{1}{r} \ln(\text{rand})$

Distribution	Generation Method	Formula/Algorithm
Arbitrary	Rejection Method	Accept x if $\text{rand} < f(x)$

1.9.2 Important Concepts

1. **Pseudorandom Numbers:** Generated by deterministic algorithms, appear random but are predictable with known seed
2. **Seed:** Initial value that determines the entire sequence (critical for reproducibility)
3. **Monte Carlo Simulation:** Probabilistic model involving an element of chance
4. **Distribution:** Describes the frequency of each outcome over many trials
5. **Multiplicative Linear Congruential Method:** $\text{random}_n = (\text{multiplier} \times \text{random}_{n-1}) \bmod \text{modulus}$

1.9.3 When to Use Computer Simulations

Use when: - Actual experiments are **not feasible** or the system **doesn't exist** - **Cost** (money, time, danger) of real experiments is **prohibitive** - Need to **test multiple alternatives** efficiently

Disadvantages: - The simulation may be **expensive** in time or money to develop - Impossible to test every alternative → can provide **good solutions but not necessarily the best** - Results may be **difficult to verify** because often we do not have real-world data - Cannot be sure we **understand what the simulation actually does** - When probabilistic with element of chance, we should be **careful of our conclusions**

1.10 Reference

A.B. Shiflet, G.W. Shiflet, *Introduction to Computational Science*, 2nd Edition, 2014, Princeton University Press.